

Evolution of the Voice-Based Appointment Booking Assistant

From an LLM-Driven Architecture to a Hybrid Deterministic Architecture

Contents

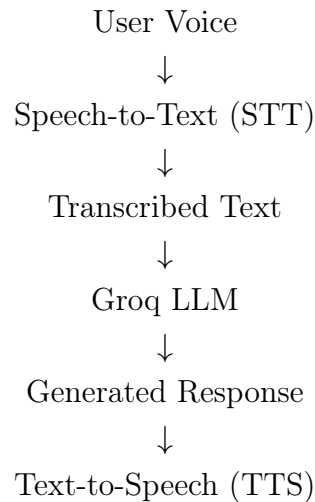
1	Original Architecture	3
1.1	Architecture Overview	3
2	Limitations of the Original Architecture	3
2.1	LLM-Based Memory	3
2.2	LLM-Controlled Workflow	4
2.3	Date Interpretation Errors	4
2.4	Doctor Assignment Errors	5
2.5	Unrestricted Response Generation	5
3	Enhanced Architecture	5
3.1	Architecture Overview	5
4	Components of the Enhanced Architecture	6
4.1	Speech-to-Text (STT)	6
4.2	Input Normalization	6
4.3	Intent Classification	6
4.4	Rule-Based Intent Overrides	7
4.5	Entity Extraction	7
4.6	Date Normalization Engine	8
4.7	Weekday Resolution Engine	8
4.8	Session State Manager	8
4.9	Validation Layer	9
4.10	Booking State Machine	9
4.11	Doctor Assignment Engine	9
4.12	Confirmation Engine	10
4.13	Post-Confirmation Query Engine	10

4.14	Response Generator	10
4.15	Text-to-Speech (TTS)	11
5	Transformation of the AI's Role	11
5.1	Original System	11
5.2	Enhanced System	11
6	Key Architectural Transformation	12

1 Original Architecture

The initial implementation followed a straightforward AI-driven architecture where the Large Language Model (LLM) was responsible for almost every aspect of the conversation.

1.1 Architecture Overview



In this design, the LLM was expected to:

- Understand user requests
- Remember previous conversation history
- Determine the next question to ask
- Extract booking information
- Generate responses
- Manage appointment flow
- Handle confirmations and cancellations

The model effectively controlled the entire system.

2 Limitations of the Original Architecture

2.1 LLM-Based Memory

The model was responsible for remembering all previously collected information.

Example:

Receptionist: What is your name?

User: Rahul

Receptionist: Which department?

User: Orthopedic

The model had to remember both the patient's name and department. Since memory existed only within the prompt context, the model could occasionally forget previously collected information and ask for it again.

Hallucination Source:

The model itself acted as the memory layer.

2.2 LLM-Controlled Workflow

The model was responsible for determining which question should be asked next.

Example:

User: I need an appointment.

AI: What time would you like?

In this case, mandatory information such as patient name, department, and date was skipped.

Hallucination Source:

The workflow existed only within the model's reasoning process.

2.3 Date Interpretation Errors

When users provided relative dates such as:

- Today
- Tomorrow
- Monday
- Friday

the model independently interpreted these expressions and could produce inconsistent results.

Hallucination Source:

Date calculation was delegated entirely to the LLM.

2.4 Doctor Assignment Errors

The model occasionally generated doctor names that did not exist within the clinic schedule.

Hallucination Source:

Information was generated instead of retrieved from actual clinic data.

2.5 Unrestricted Response Generation

Every response was dynamically generated by the model, including simple workflow messages such as:

“Please tell me your name.”

This introduced unnecessary variability into the system.

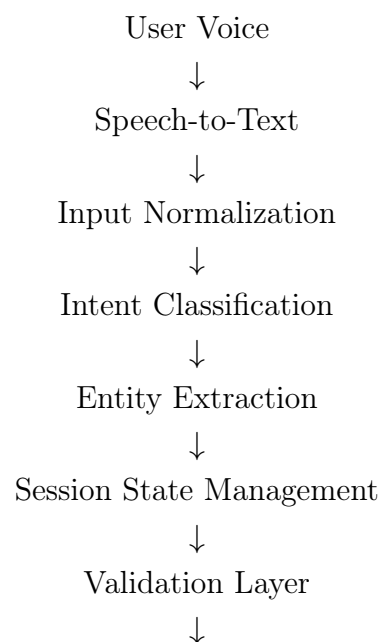
Hallucination Source:

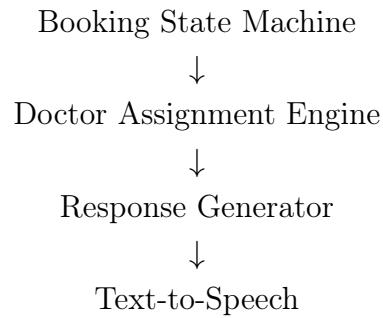
The model possessed excessive freedom in response generation.

3 Enhanced Architecture

To overcome the shortcomings of the original design, the architecture was redesigned into a hybrid system where deterministic software components govern the booking workflow while the LLM performs only restricted tasks.

3.1 Architecture Overview





The core principle of the redesigned architecture is:

The AI extracts information, while the system makes decisions.

4 Components of the Enhanced Architecture

4.1 Speech-to-Text (STT)

Responsibility

Convert user speech into text.

Role in Hallucination Reduction

This component does not directly reduce hallucinations. Its purpose is limited to accurate transcription.

4.2 Input Normalization

Responsibility

Standardize multilingual inputs into canonical forms.

Examples:

- Bengali date references converted into Hindi equivalents
- Department synonyms mapped to canonical department names

Role in Hallucination Reduction

Reduces ambiguity caused by linguistic variations and provides consistent inputs to downstream AI components.

4.3 Intent Classification

Responsibility

Identify the user's intent.

Supported intents include:

- PROVIDE_INFO
- ASK_QUERY
- CONTINUE
- END
- CANCEL
- RESCHEDULE

Role in Hallucination Reduction

The model no longer controls the conversation flow. It only classifies user intent.

4.4 Rule-Based Intent Overrides

Responsibility

Validate and override incorrect intent classifications.

Example:

If the user asks:

“When is my appointment?”

the system directly classifies the request as an appointment query even if the model predicts another intent.

Role in Hallucination Reduction

Critical workflow decisions do not depend entirely on AI predictions.

4.5 Entity Extraction

Responsibility

Extract structured information from user input.

Example:

```
{  
  "name": "Rahul",  
  "department": "Orthopedic",  
  "date": "2026-06-08",  
  "time": "10 AM"  
}
```

Role in Hallucination Reduction

The model is restricted to extracting information rather than generating workflow decisions.

4.6 Date Normalization Engine

Responsibility

Create deterministic mappings for relative date expressions.

Examples:

- Today = 2026-06-05
- Tomorrow = 2026-06-06
- Monday = 2026-06-08

Role in Hallucination Reduction

The model no longer performs date calculations independently.

4.7 Weekday Resolution Engine

Responsibility

Convert weekday references directly into actual calendar dates.

Role in Hallucination Reduction

Eliminates ambiguity in weekday interpretation.

4.8 Session State Manager

Responsibility

Maintain conversation memory.

Stored information includes:

- Patient Name
- Department
- Date
- Time
- Assigned Doctor
- Conversation Mode

Role in Hallucination Reduction

Memory is maintained by software rather than the LLM, preventing information loss.

4.9 Validation Layer**Responsibility**

Validate extracted entities before storing them.

Examples:

- Invalid names are rejected
- Invalid time formats are rejected
- Past dates are rejected

Role in Hallucination Reduction

Incorrect model outputs are filtered before entering the booking workflow.

4.10 Booking State Machine**Responsibility**

Control the appointment booking workflow.

Example logic:

- If name is missing → Ask Name
- If department is missing → Ask Department
- If date is missing → Ask Date
- If time is missing → Ask Time

Role in Hallucination Reduction

Conversation flow becomes deterministic and cannot skip mandatory steps.

4.11 Doctor Assignment Engine**Responsibility**

Assign doctors using predefined clinic schedules.

Inputs:

- Department
- Appointment Date

Output:

- Assigned Doctor

Role in Hallucination Reduction

Doctor information is retrieved from clinic data rather than generated by the model.

4.12 Confirmation Engine

Responsibility

Verify appointment details before final confirmation.

Role in Hallucination Reduction

Prevents incomplete appointments from being confirmed.

4.13 Post-Confirmation Query Engine

Responsibility

Answer booking-related questions after confirmation.

Examples:

- Appointment date
- Appointment time
- Assigned doctor
- Department

Role in Hallucination Reduction

Responses are generated directly from stored appointment data.

4.14 Response Generator

Responsibility

Generate controlled workflow responses.

Examples:

- Ask Name
- Ask Department
- Ask Date
- Ask Time

- Cancel Appointment
- End Conversation

Role in Hallucination Reduction

Critical workflow responses are deterministic and not generated dynamically by the LLM.

4.15 Text-to-Speech (TTS)

Responsibility

Convert the final response into speech.

Role in Hallucination Reduction

This component does not directly influence hallucination reduction.

5 Transformation of the AI's Role

5.1 Original System

The LLM acted as:

- Memory Manager
- Workflow Controller
- Decision Maker
- Date Interpreter
- Doctor Selector
- Response Generator

All system behavior depended on AI reasoning.

5.2 Enhanced System

The LLM is restricted to:

- Intent Classification
- Information Extraction
- Limited Response Formatting

All critical business decisions are handled through deterministic software components.

6 Key Architectural Transformation

The most significant change in the system is the shift of control.

Original Architecture

AI $\rightarrow$ Controls System

Enhanced Architecture

System $\rightarrow$ Controls AI

The appointment booking workflow no longer depends on what the model believes should happen next. Dedicated software components now manage memory, validation, scheduling, doctor assignment, confirmation, and conversation flow, while the AI is restricted to extracting structured information from natural language.

This separation of responsibilities is the primary reason for the substantial reduction in hallucinations and the improved reliability of the enhanced architecture.